

Ragionato per 12s

TSIA2 Math Practice Test 10

PART A — QUESTIONS

TSIA2-M10-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A copy center charges \$0.12 per page for the first 80 pages and \$0.08 per page for each additional page. What is the total cost to print 125 pages?

A) \$10.00

B) \$12.40

C) \$13.20

D) \$15.00

TSIA2-M10-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for x: $(x/3) + 7 = 15$

A) 8

B) 18

C) 22

D) 24

TSIA2-M10-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular swimming area is 25 meters long and 14 meters wide. A safety rope is placed around the entire perimeter. How many meters of rope are needed?

A) 39 meters

B) 64 meters

C) 78 meters

D) 350 meters

TSIA2-M10-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A student recorded the number of minutes spent reading on six nights:

35, 40, 42, 45, 48, 60

What is the median number of minutes?

- A) 42
- B) 43.5
- C) 45
- D) 48

TSIA2-M10-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A storage container holds 18 boxes. Each box weighs 12.5 pounds. What is the total weight of the boxes?

- A) 205 pounds
- B) 215 pounds
- C) 225 pounds
- D) 235 pounds

TSIA2-M10-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A line passes through the points (1, 6) and (5, 18). What is the slope of the line?

- A) 2
- B) 3
- C) 4
- D) 12

TSIA2-M10-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A parallelogram has a base of 13 centimeters and a height of 9 centimeters. What is its area?

- A) 22 square centimeters
- B) 44 square centimeters
- C) 117 square centimeters
- D) 234 square centimeters

TSIA2-M10-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A drawer contains 3 AA batteries, 5 AAA batteries, and 4 rechargeable batteries. One battery is selected at random. What is the probability that the battery is AAA?

- A) $\frac{1}{4}$
- B) $\frac{5}{12}$
- C) $\frac{7}{12}$
- D) $\frac{5}{7}$

TSIA2-M10-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student answers 42 questions correctly on a 56-question practice test. What percent of the questions did the student answer correctly?

A) 70%

B) 72%

C) 75%

D) 80%

TSIA2-M10-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A savings plan starts with \$120 and increases by \$35 each month. If m represents the number of months and S represents the total savings in dollars, which equation models this situation?

A) $S = 120m + 35$

B) $S = 35m + 120$

C) $S = 155m$

D) $S = 35(m + 120)$

TSIA2-M10-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular clock face has a radius of 7 inches. What is the area of the clock face?

A) 14π square inches

B) 28π square inches

C) 49π square inches

D) 98π square inches

TSIA2-M10-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The table shows the number of applications processed by an office during four days.

Monday: 52

Tuesday: 47

Wednesday: 61

Thursday: 56

What is the mean number of applications processed per day?

A) 52

B) 54

- C) 56
- D) 58

TSIA2-M10-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A part-time worker earns \$14.25 per hour. If the worker earns \$256.50 before taxes, how many hours did the worker work?

- A) 16 hours
- B) 17 hours
- C) 18 hours
- D) 19 hours

TSIA2-M10-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $4x^2 - 25$?

- A) $(2x - 5)(2x + 5)$
- B) $(4x - 5)(x + 5)$
- C) $(2x - 25)(2x + 1)$
- D) $(4x - 25)(x + 1)$

TSIA2-M10-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A ladder reaches a window 16 feet above the ground. The base of the ladder is 12 feet from the wall. What is the length of the ladder?

- A) 20 feet
- B) 24 feet
- C) 28 feet
- D) 32 feet

TSIA2-M10-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A school club has 8 freshmen, 6 sophomores, and 4 juniors. If one member is selected at random, what is the probability that the member is not a freshman?

- A) $\frac{4}{9}$
- B) $\frac{5}{9}$
- C) $\frac{2}{3}$
- D) $\frac{3}{4}$

TSIA2-M10-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A bag of soil weighs 37.8 pounds. Which whole number is closest to $\sqrt{37.8}$?

- A) 5
- B) 6
- C) 7
- D) 8

TSIA2-M10-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve the inequality $5x + 6 \leq 31$.

- A) $x \leq 5$
- B) $x \geq 5$
- C) $x \leq 25$
- D) $x \geq 25$

TSIA2-M10-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A model bridge is 18 inches long. The actual bridge is 270 feet long. If 1 inch on the model represents the same number of feet on the actual bridge, what does 1 inch represent?

- A) 12 feet
- B) 15 feet
- C) 18 feet
- D) 30 feet

TSIA2-M10-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A data set contains the values 9, 12, 15, 18, and 21. If each value in the data set is increased by 4, what happens to the mean?

- A) The mean increases by 4.
- B) The mean decreases by 4.
- C) The mean is multiplied by 4.
- D) The mean stays the same.

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M10-001 — Correct: C — Correct Answer: \$13.20 — Explanation: The first 80 pages cost $80 \times \$0.12 = \9.60 . The remaining 45 pages cost $45 \times \$0.08 = \3.60 . The total cost is \$13.20.

TSIA2-M10-002 — Correct: D — Correct Answer: 24 — Explanation: Subtract 7 from both sides

to get $x/3 = 8$. Multiply both sides by 3 to get $x = 24$.

TSIA2-M10-003 — Correct: C — Correct Answer: 78 meters — Explanation: The perimeter of the rectangle is $2(25 + 14) = 78$ meters.

TSIA2-M10-004 — Correct: B — Correct Answer: 43.5 — Explanation: With six values, the median is the average of the third and fourth values: $(42 + 45) \div 2 = 43.5$.

TSIA2-M10-005 — Correct: C — Correct Answer: 225 pounds — Explanation: Multiply the number of boxes by the weight per box: $18 \times 12.5 = 225$ pounds.

TSIA2-M10-006 — Correct: B — Correct Answer: 3 — Explanation: The slope is the change in y divided by the change in x: $(18 - 6)/(5 - 1) = 12/4 = 3$.

TSIA2-M10-007 — Correct: C — Correct Answer: 117 square centimeters — Explanation: The area of a parallelogram is base $\times$ height, so $13 \times 9 = 117$ square centimeters.

TSIA2-M10-008 — Correct: B — Correct Answer: 5/12 — Explanation: There are $3 + 5 + 4 = 12$ batteries total, and 5 are AAA. The probability is 5/12.

TSIA2-M10-009 — Correct: C — Correct Answer: 75% — Explanation: The fraction correct is $42/56 = 3/4$, which equals 75%.

TSIA2-M10-010 — Correct: B — Correct Answer: $S = 35m + 120$ — Explanation: The savings increase by \$35 each month and start at \$120, so $S = 35m + 120$.

TSIA2-M10-011 — Correct: C — Correct Answer: 49π square inches — Explanation: The area of a circle is $A = \pi r^2$. With $r = 7$, $A = 49\pi$ square inches.

TSIA2-M10-012 — Correct: B — Correct Answer: 54 — Explanation: The total is $52 + 47 + 61 + 56 = 216$. Dividing by 4 gives a mean of 54.

TSIA2-M10-013 — Correct: C — Correct Answer: 18 hours — Explanation: Divide total earnings by hourly pay: $\$256.50 \div \$14.25 = 18$ hours.

TSIA2-M10-014 — Correct: A — Correct Answer: $(2x - 5)(2x + 5)$ — Explanation: This is a difference of squares: $4x^2 - 25 = (2x)^2 - 5^2 = (2x - 5)(2x + 5)$.

TSIA2-M10-015 — Correct: A — Correct Answer: 20 feet — Explanation: The ladder is the hypotenuse of a right triangle. Its length is $\sqrt{(16^2 + 12^2)} = \sqrt{400} = 20$ feet.

TSIA2-M10-016 — Correct: B — Correct Answer: 5/9 — Explanation: There are 18 club members total. Members who are not freshmen total $6 + 4 = 10$, so the probability is $10/18 = 5/9$.

TSIA2-M10-017 — Correct: B — Correct Answer: 6 — Explanation: Since $6^2 = 36$ and $7^2 = 49$, $\sqrt{37.8}$ is between 6 and 7 and closer to 6.

TSIA2-M10-018 — Correct: A — Correct Answer: $x \leq 5$ — Explanation: Subtract 6 from both sides to get $5x \leq 25$. Divide by 5 to get $x \leq 5$.

TSIA2-M10-019 — Correct: B — Correct Answer: 15 feet — Explanation: Divide the actual bridge length by the model length: $270 \div 18 = 15$ feet per inch.

TSIA2-M10-020 — Correct: A — Correct Answer: The mean increases by 4. — Explanation: When the same number is added to every value in a data set, the mean increases by that same number.